

# Analysis Problem Set #2

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## Q1

Beginning from the triangle inequality,

$$|x + z| \leq |x| + |z|.$$

For  $z = x - y$ , so that  $x = y + z$ :

$$|x| \leq |y| + |x - y| \implies |x| - |y| \leq |x - y|.$$

Since  $x$  and  $y$  are arbitrary, interchanging their labels gives

$$|y| - |x| \leq |y - x| = |x - y| \implies -|x - y| \leq |x| - |y|.$$

Taken together,

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is exactly

$$||x| - |y|| \leq |x - y|. \quad \blacksquare$$

## Q2 (version 1)<sup>1</sup>

For  $f(x) = e^x$ ,  $f^{(n)}(x) = e^x$ . For  $a = 0$ ,  $e^a = 1$ .

So the Taylor polynomial ( $n = 5$ ) is:

$$1 + (1)(x) + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{120}x^5.$$

At  $x = 1$ , this evaluates to  $\approx e$ .

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<sup>1</sup>Probably want to choose a version for Q2

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## Q2 (version 2)

$$\begin{aligned} T_a(x) &= e^a + e^a(x-a) + \frac{e^a}{2!}(x-a)^2 + \frac{e^a}{3!}(x-a)^3 + \frac{e^a}{4!}(x-a)^4 + \frac{e^a}{5!}(x-a)^5 \\ T_0(x) &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} \\ T_0(1) &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} \\ T_0(1) &= \frac{120+120+60+20+5+1}{120} \\ T_0(1) &= \frac{326}{120} \end{aligned}$$

## Q3

By the chain rule,

$$\frac{d \log(x(t))}{dt} = \frac{1}{x} \frac{dx(t)}{dt} = \frac{\frac{dx(t)}{dt}}{x}$$

## Q4

### 4.1

$$\begin{aligned} H_f &= \begin{bmatrix} -\frac{1}{4}x^{-3/2} & 0 \\ 0 & -\frac{1}{4}y^{-3/2} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -\frac{1}{x^{3/2}} & 0 \\ 0 & -\frac{1}{y^{3/2}} \end{bmatrix} \\ |Hf| &= \frac{1}{16} \cdot \frac{1}{(xy)^{3/2}} > 0 \end{aligned}$$

The Hessian is negative definite, so  $f$  is strictly concave over  $\mathbb{R}_{++}^2$ , its domain.

**4.2**  $g(x, y) = \sqrt{xy}$

$$Hg = \begin{bmatrix} -\frac{\sqrt{y}}{4} x^{-3/2} & \frac{1}{4\sqrt{xy}} \\ \frac{1}{4\sqrt{xy}} & -\frac{\sqrt{x}}{4} y^{-3/2} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -\frac{y^{1/2}}{x^{3/2}} & \frac{1}{\sqrt{xy}} \\ \frac{1}{\sqrt{xy}} & -\frac{x^{1/2}}{y^{3/2}} \end{bmatrix}$$

$$|Hg| = \frac{1}{xy} - \frac{1}{xy} = 0$$

The Hessian is negative semidefinite, so  $g$  is concave but not strictly concave over  $\mathbb{R}_{++}^2$ , its domain.

## Q5

### 5.1

For any  $n$ ,  $I_n$  is an open set. Choose any  $x \in (-\frac{1}{n}, \frac{1}{n})$ . Let  $0 < r < \frac{1}{n}$ . Then  $B(x, r) \subset I_n$ .

$$\bigcap_{n=1}^{\infty} I_n = \{0\},$$

because for any  $z \in \mathbb{R}$ ,  $z \neq 0$ , there exists  $n \in \mathbb{N}$  such that

$$d(0, \frac{1}{n}) < d(0, z) \iff \left| \frac{1}{n} \right| < |z| \implies z \notin I_n.$$

And  $0 \in I_n$  because  $0 < \frac{1}{n}$  for every  $n \in \mathbb{N}$ .

$\{0\}$  is a closed set because it is the complement of  $(-\infty, 0) \cup (0, \infty)$ , which is the union of two open balls and thus an open set by way of Theorems 1 and 2.

### 5.2

For any  $n$ ,  $I_n$  is a closed set. Choose  $x = \frac{1}{n+1} \in I_n$ . Then for all  $r > 0$ ,  $B(x, r)$  is not a subset of  $J_n$

$$\text{For center } x = \frac{1}{2} \text{ and radius } r = \frac{n-1}{n+1},$$

$$\left( \frac{1}{2} - \frac{1}{n+1}, \frac{n-1}{n+1} \right)$$

we have  $\forall y \in J_n$ ,  $d(x, y) \leq r$ . Also, the complement of  $J_n$  is the open set

$$\left( -\infty, \frac{1}{n+1} \right) \cup \left( \frac{n}{n+1}, \infty \right).$$

Taken together, that implies that  $J_n$  is a closed ball.

For  $\bigcup_{i=1}^{\infty} J_n$ :

$0 < \frac{1}{n+1}$  for any  $n$ , so for any  $u \leq 0$ ,  $u \notin \bigcup_{i=1}^{\infty} J_n$ .

Similarly  $1 > 1 - \frac{1}{n+1}$  for any  $n$ , so for any  $v \geq 1$ ,  $v \notin \bigcup_{i=1}^{\infty} J_n$ .

**Q6**

**Q7**

Bordered Hessian:

$$H = \begin{bmatrix} 0 & \alpha AK^{\alpha-1} L^{1-\alpha} & (1-\alpha) AK^{\alpha} L^{-\alpha} \\ \alpha AK^{\alpha-1} L^{1-\alpha} & (\alpha^2 - \alpha) AK^{\alpha-2} L^{1-\alpha} & (\alpha - \alpha^2) AK^{\alpha-1} L^{-\alpha} \\ (1-\alpha) AK^{\alpha} L^{-\alpha} & (\alpha - \alpha^2) AK^{\alpha-1} L^{-\alpha} & (\alpha^2 - \alpha) AK^{\alpha} L^{-\alpha-1} \end{bmatrix}$$

where

$$\begin{aligned} a &= \alpha AK^{\alpha-1} L^{1-\alpha}, & b &= (1-\alpha) AK^{\alpha} L^{-\alpha}, & c &= (\alpha^2 - \alpha) AK^{\alpha-2} L^{1-\alpha}, \\ d &= (\alpha - \alpha^2) AK^{\alpha-1} L^{-\alpha}, & e &= (\alpha^2 - \alpha) AK^{\alpha} L^{-\alpha-1}. \end{aligned}$$

**Leading principal minors**

$$|H_2| = -\alpha^2 A^2 K^{2\alpha-2} L^{2-2\alpha} \leq 0 \quad \text{for } K, L \geq 0$$

$$\begin{aligned} |H_3| &= 2abd - a^2e - b^2c \\ &= 2(\alpha)(1-\alpha)(\alpha - \alpha^2) A^3 K^{3\alpha-2} L^{1-3\alpha} - (\alpha^4 - \alpha^3) A^3 K^{3\alpha-2} L^{1-3\alpha} \\ &\quad - (1-\alpha)^2 (\alpha^2 - \alpha) A^3 K^{3\alpha-2} L^{1-3\alpha}, \end{aligned}$$

need to show  $\geq 0$  for  $K, L \geq 0$ .

Divide by  $(1-\alpha)A^3$ :

$$2\alpha^2(1-\alpha) \cancel{K^{3\alpha-2} L^{1-3\alpha}} \geq -\alpha^3 \cancel{K^{3\alpha-2} L^{1-3\alpha}} + (1-\alpha)(\alpha^2 - \alpha) \cancel{K^{3\alpha-2} L^{1-3\alpha}}$$

$$2\alpha^2 \geq (1-\alpha)(\alpha^2 - \alpha) - \alpha^3$$

$$2\alpha \geq (1-\alpha)(\alpha - 1) - \alpha^2$$

$$2\alpha \geq -((\alpha - 1)^2 + \alpha^2)$$

$$2\alpha \geq -(2\alpha^2 - 2\alpha + 1)$$

$$0 \geq -2\alpha^2 - 1 \quad \checkmark$$

**Q8**

**Q9**

1.  $A$  is bounded,  $M = \sqrt{10}$ ,  $\bar{x} = 0$ .
2.  $B$  is not bounded. All  $x < 0$  are in  $B$ .
3.  $C$  is not bounded. Assuming  $\mathbb{R}_+$  means non-negative, all  $(0, y)$  and  $(x, 0)$  are in  $C$ .
4.  $D$  is bounded,  $M = 10$  and  $\bar{x} = (0, 0)$ .